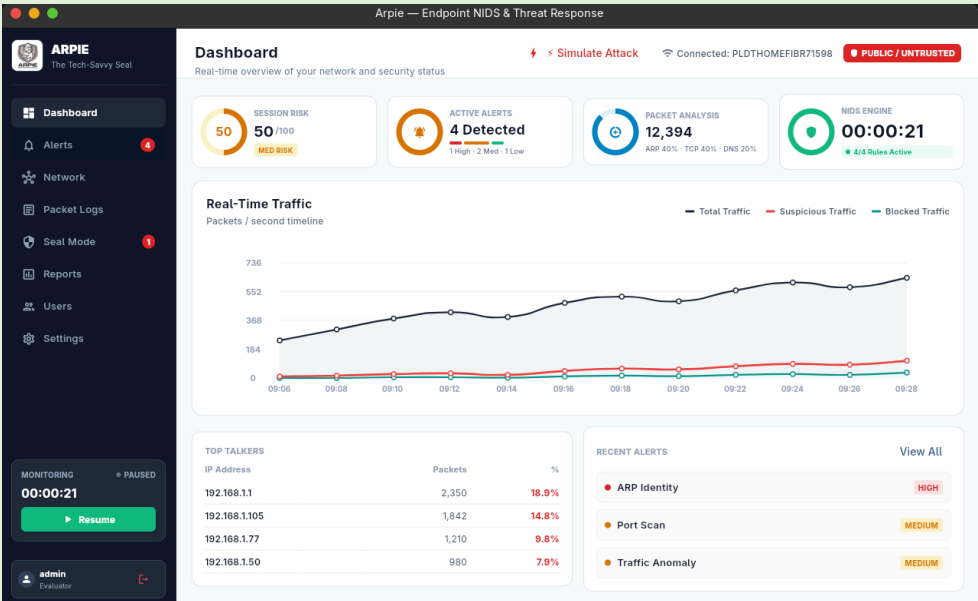


User Interface and System Flow -1 st Hands-on Exam	
NAME	ERA DUMANGCAS
PROJECT TITLE (specific)	Arpie: A Context-Aware Endpoint Network Intrusion Detection and Threat-Response System for Public Wi-Fi
SUBJECT: CODE: TIME: Date of Project Presentation:	INFORMATION ASSURANCE AND SECURITY 2 5938 8:00-10:00AM
PROJECT NAME	#4 Network Intrusion Detection System
Technologies and Framework:	Programming Language - Python Network Analysis and Security Libraries - Scapy (Python) - psutil (Python) - iptables Application Framework - Flet (Python) - notify-send Threat Intelligence - AbuseIPDB (API) - IPinfo Lite (API) - ReportLab (Python) Database - SQLite Testings - pytest (Python) - PCAP Files
Detection Techniques	1. ARP Identity Inconsistency One local IP associated with more than one MAC address within a 5-minute window 2. Port-Scan Behavior More than 15 unique destination ports contacted by one source within a 10-second window 3. Traffic-Rate Anomaly SYN/UDP/ICMP packets from one source exceeding 100 packets/second (default baseline, configurable) 4. Gateway Identity Change gateway MAC/IP association changes more than once within a 10-minute window

Required Data	1. Security Events & Alerts 2. Incident Evidence Details 3. Threat Intelligence 4. User Account Information 5. Network Context & Devices 6. Monitoring Session Details 7. Packet Logs 8. Seal Actions 9. System Configuration
Use Diagram	<pre> graph LR subgraph Arpie ["Arpie: Endpoint NIDS & Threat Response"] subgraph Auth ["Authentication & Profile"] A1([Authenticate & Sign In]) A2([Initialize Setup Wizard]) A3([Manage User Accounts & Roles]) end subgraph Triage ["Threat Triage & Forensics"] T1([Check Threat Reputation]) T2([Inspect Evidence Details]) T3([View Security Alerts]) T4([Export Forensic Reports]) end subgraph Config ["Containment & System Configuration"] C1([Configure Detection Thresholds]) C2([Manage Threat Intel API Keys]) C3([Isolate Host Seal Mode]) end subgraph Telemetry ["Real-Time Telemetry & Monitoring"] Te1([View Real-Time Dashboard]) Te2([View Network Context & Devices]) Te3([View Packet Capture Logs]) end end EndUser((End User)) EvalAdmin((Evaluator / Administrator)) EndUser --> A1 EndUser --> A2 EndUser --> T3 EndUser --> Te2 EndUser --> Te3 EvalAdmin --> A2 EvalAdmin --> A3 EvalAdmin --> T1 EvalAdmin --> T2 EvalAdmin --> T3 EvalAdmin --> T4 EvalAdmin --> C1 EvalAdmin --> C2 EvalAdmin --> C3 EvalAdmin --> Te1 EvalAdmin --> Te2 EvalAdmin --> Te3 EvalAdmin -.-> EndUser T1 -.-> «Service» ETI[External Threat Intel AbuseIPDB] C3 -.-> «System» OFS[OS Firewall Subsystem] </pre>
Target Users	• Students and Academic Researchers • Remote and Hybrid Professionals • Freelancers and Small-Office/Home-Office (SOHO) Users • Academic Evaluators and Cybersecurity Auditors
User Roles	• Evaluator / Administrator ○ Configure detection thresholds and API credentials. ○ Replay offline PCAP attack captures for testing. ○ Simulate demonstration threats and calibrate rules. ○ Inspect all alerts, packet logs, and forensic reports.

- End User
 - Monitor real-time network security and risk score.
 - Review explainable alerts and evidence details.
 - Apply and release 1-click Seal Mode host containment.
 - View active packet streams and export session reports.

Real-Time Threat Monitoring Dashboard



Description:
Shows a live overview of network activity and security status. It displays current traffic speed, the overall session risk score, detected attack types, and the most active devices on the network.

Security Events & Alerts

Alerts
All detected security events for this monitoring session

4 Active Alerts

Search alerts by IP, threat type or evidence proof...

Security Events & Alerts
4 Incidents Logged

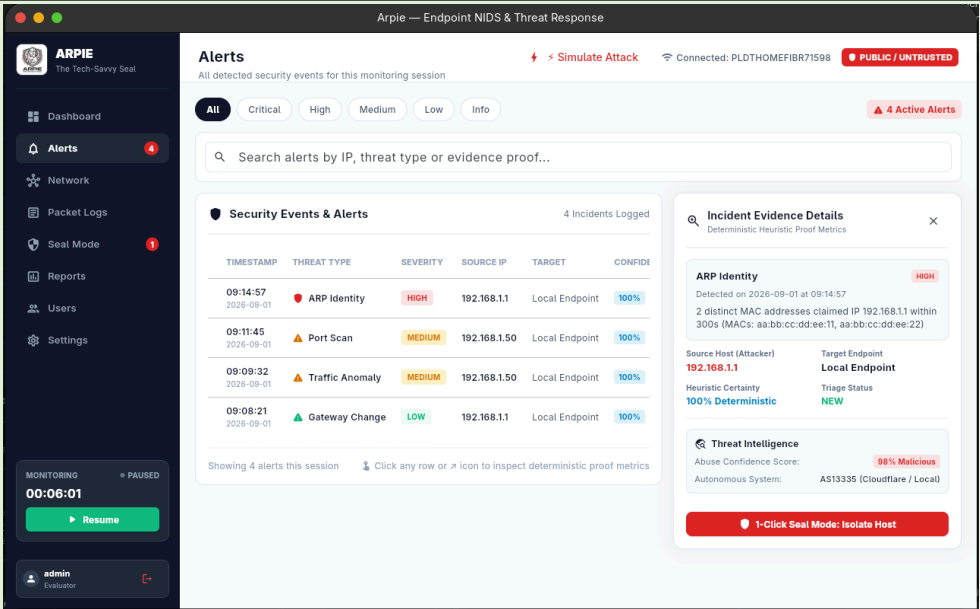
TIMESTAMP	THREAT TYPE	SEVERITY	SOURCE IP	TARGET	CONFIDENCE	STATUS	ACTION
09:14:57 2028-09-01	● ARP Identity	HIGH	192.168.1.1	Local Endpoint	100%	NEW	🔍
09:11:45 2028-09-01	▲ Port Scan	MEDIUM	192.168.1.50	Local Endpoint	100%	NEW	🔍
09:09:32 2028-09-01	▲ Traffic Anomaly	MEDIUM	192.168.1.50	Local Endpoint	100%	ACK	🔍
09:08:21 2028-09-01	▲ Gateway Change	LOW	192.168.1.1	Local Endpoint	100%	REVIEW	🔍

Showing 4 alerts this session

Click any row or icon to inspect deterministic proof metrics

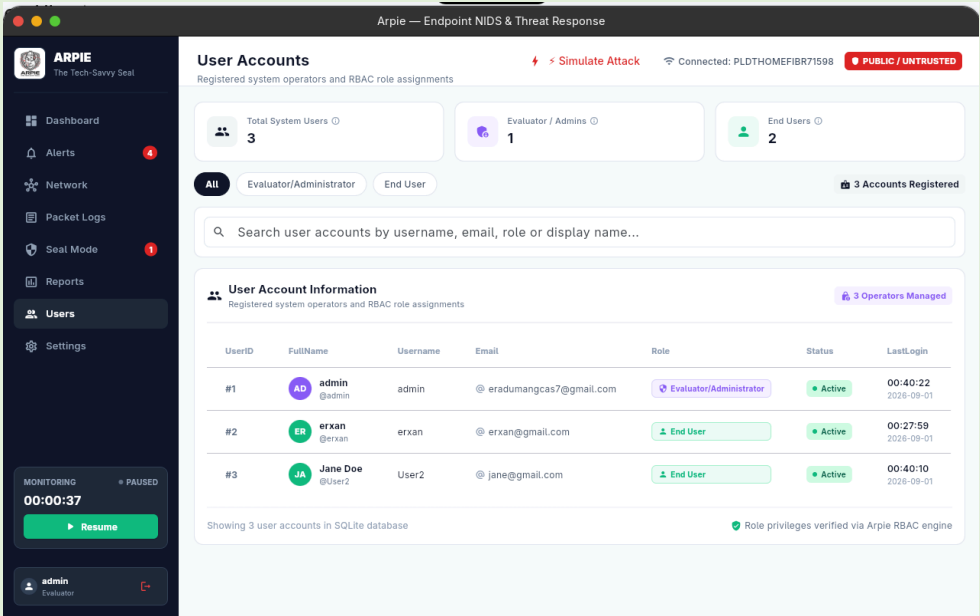
Description:
Displays a table of all detected security threats during the session. It lists the detection time, threat type (such as ARP spoofing or port scans), severity level, attacker IP, target device, and alert status.

Incident Evidence
Details & Threat
Intelligence



Description:
A side panel that opens when an alert is clicked to show detailed proof of the attack. It displays technical evidence (such as conflicting MAC addresses or scanned ports), external reputation data (AbuseIPDB score and country), and a button to quickly isolate the attacker.

User Account
Information



Description:
A table of all registered accounts in the system. It lists user IDs, full names, usernames, email addresses, assigned roles (Evaluator/Administrator or End User), account statuses, and last login times.

Network Context & Devices

ARPIE

The Tech-Savvy Seal

Dashboard

Alerts

Network

Packet Logs

Seal Mode

Reports

Users

Settings

MONITORING

PAUSED

00:00:01

Resume

admin

Evaluator

Arpie — Endpoint NIDS & Threat Response

Network

Network context assessment and discovered device inventory

Network Context Assessment

Environmental trust evaluation, gateway bindings, and subnet parameters.

SSID / NETWORK NAME

PLDTHOMEFIBR71598

802.11 Wi-Fi Infrastructure

GATEWAY IP & MAC

192.168.1.1 (00:50:56:C0:00:01)

Validated Default Gateway Binding

SUBNET CIDR

192.168.1.0/24

Local Broadcast Domain Scope

INTERFACE & SENSITIVITY

wlan0 (Promiscuous)

Elevated Heuristic Sensitivity

Network Device Inventory

Tracking 5 discovered active endpoints on subnet

Scan Local Subnet

DeviceID	Hostname	IP Address	MAC Address	Vendor	Device Type	Status	Last
#1	Gateway-Router (AP)	192.168.1.1	00:50:56:C0:00:01	Cisco / Fiberhome	Gateway / Router	Active Gateway	202
#2	This-Endpoint (Local Host)	192.168.1.100	00:0C:29:4F:11:AA	Intel Corp.	Endpoint / Host	Local Endpoint	202
#3	Laptop-Work	192.168.1.15	00:1E:50:32:8B:CC	Dell Inc.	Laptop	Known Host	202
#4	Office-Printer	192.168.1.20	00:1B:44:12:34:7F	HP Inc.	Printer / IoT	Known Host	202
#5	Suspect-Node	192.168.1.50	3C:97:0E:12:34:56	Unknown (Attacker)	Workstation	Untrusted / Suspect	202

Description:
Shows information about the connected network and all active devices. It displays the Wi-Fi name, gateway IP/MAC address, and network trust level, along with a table of all discovered devices on the local network.

Monitoring Session Details

ARPIE

The Tech-Savvy Seal

Dashboard

Alerts

Network

Packet Logs

Seal Mode

Reports

Users

Settings

MONITORING

PAUSED

00:00:40

Resume

admin

Evaluator

Arpie — Endpoint NIDS & Threat Response

Forensic Reports

Automated incident reports, threat summaries, and exports

Forensic Incident Reports & Session Exports

Logged in as: admin (admin) · Historical sessions and evidence logs

Live sniffing notice: [Errno 1] Operation not permitted

Export Active PDF

Export Active HTML

Export Active JSON

Active Session Executive Summary

Session ID: #33 (Current Active)

Network: PLDTHOMEFIBR71598 (PUBLIC / UNTRUSTED)

Detected Threats: 4 security incidents

Overall Risk Index: 50 / 100 (HIGH RISK)

Past Monitoring Sessions & Audit History

Retrieve and re-export past session reports recorded for your account.

SESSION	DATE / TIME	NETWORK	THREATS	EXPORT
#33	2026-09-01 00:56	PLDTHOMEFIBR71598 (PUBLIC-UNTRUSTED)	0 events	
#32	2026-09-01 00:54	PLDTHOMEFIBR71598 (PUBLIC-UNTRUSTED)	0 events	

Description:
Displays the details of the active monitoring session, including the session ID, network interface, operator name, start and end times, total duration, and overall risk rating.

Packet Logs

ARPIE

The Tech-Savvy Seal

Dashboard

Alerts

Network

Packet Logs

Seal Mode

Reports

Users

Settings

MONITORING

00:02:14

Resume

admin

Evaluator

Arpie — Endpoint NIDS & Threat Response

Packet Capture Logs

Live inspection and recorded packet stream replay

Offline PCAP Attack Replay (Evaluator Tool)
sample_pcaps/demo_attack.pcap

Replay PCAP File

Timestamp	Source IP	Dest IP	Protocol	Source MAC	Dest MAC	Length
00:58:54.479	192.168.1.1	192.168.1.15	UDP (NTP)	00:50:56:C0:00:01	00:1E:50:32:8B:CC	90
00:58:53.478	192.168.1.20	192.168.1.1	TCP ACK (HTTPS)	00:1B:44:12:34:7F	00:50:56:C0:00:01	1460
00:58:52.477	192.168.1.1	192.168.1.15	UDP (NTP)	00:50:56:C0:00:01	00:1E:50:32:8B:CC	90
00:58:51.477	192.168.1.1	192.168.1.15	UDP (NTP)	00:50:56:C0:00:01	00:1E:50:32:8B:CC	90
00:58:50.476	192.168.1.20	192.168.1.1	TCP ACK (HTTPS)	00:1B:44:12:34:7F	00:50:56:C0:00:01	1460
00:58:49.475	192.168.1.10	192.168.1.1	ARP	00:0C:29:4F:11:AA	00:50:56:C0:00:01	42
00:58:48.475	192.168.1.1	192.168.1.15	UDP (NTP)	00:50:56:C0:00:01	00:1E:50:32:8B:CC	90
00:58:47.474	192.168.1.10	192.168.1.1	ARP	00:0C:29:4F:11:AA	00:50:56:C0:00:01	42
00:58:46.473	192.168.1.105	142.250.190.46	TLSv1.3 Handshake	00:1E:50:32:8B:CC	00:50:56:C0:00:01	512
00:58:45.472	192.168.1.20	192.168.1.1	TCP ACK (HTTPS)	00:1B:44:12:34:7F	00:50:56:C0:00:01	1460
00:58:44.472	192.168.1.1	192.168.1.15	UDP (NTP)	00:50:56:C0:00:01	00:1E:50:32:8B:CC	90

Description:
Shows a live stream of captured network packets. It lists the timestamp, source and destination IP addresses, protocol (ARP, TCP, UDP, DNS), packet size, TCP flags, and basic packet information.

Seal Actions

ARPIE

The Tech-Savvy Seal

Dashboard

Alerts

Network

Packet Logs

Seal Mode

Reports

Users

Settings

MONITORING

00:06:41

Resume

admin

Evaluator

Arpie — Endpoint NIDS & Threat Response

Seal Mode Mitigation

Reversible endpoint threat response and network isolation

Endpoint Seal Mode Threat Mitigation

Automated and reversible host isolation using iptables/netsh.

Activate Emergency Seal

ACTIVE CONTAINMENT RULES

1 Blocked IPs

Host isolation is active and dropping suspicious ingress/egress.

FIREWALL STATE

Reversible & Safe

All routing tables restore automatically on session teardown.

Currently Isolated Hosts

Hostile Attacker: 192.168.1.50

Rule: DROP ALL INBOUND/OUTBOUND (iptables -A INPUT -s ... -j DROP)

Unblock

Description:
Lists all suspect devices blocked or isolated by the system. It displays the blocked IP address, reason for containment, timestamp, current block status, and a button to unblock the device.

System Configuration

Arpie — Endpoint NIDS & Threat Response

System Configuration

Heuristic thresholds, threat intel feeds, and rules

⚡ Simulate Attack Connected: PLDTHOMEFIBR71598 PUBLIC / UNTRUSTED

Account & Security Profile

Logged in as admin Evaluator/Administrator

Account Role: Evaluator/Administrator Email: eradumangcas7@gmail.com Last Login: 2026-09-01 00:56

Username: admin

Current Password New Password

Update Credentials

Detection Thresholds & Threat Intel Feeds (Evaluator Tools)

Configure heuristic sensitivity and external threat intelligence APIs.

AbuseIPDB API Key IPInfo Token

Detection Rule Thresholds

Traffic Anomaly Threshold 100 packets/sec

Port Scan Trigger Threshold 15 ports / 10 sec

ARP Identity Window 5 minutes

Gateway Change Window 10 minutes

Save Detection Settings

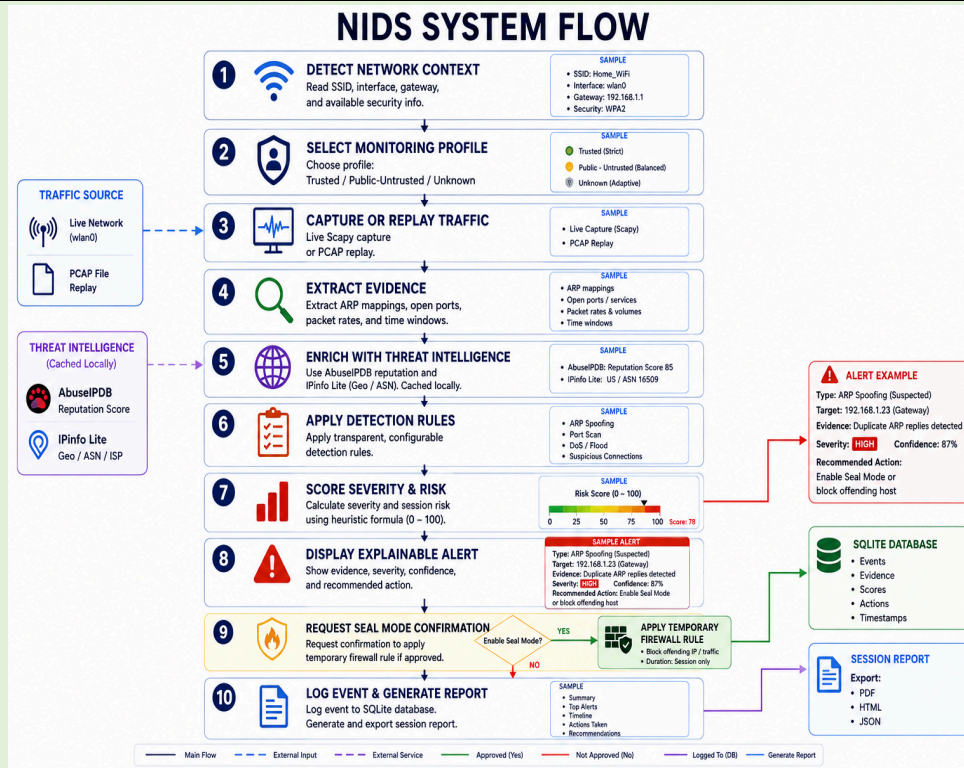
MONITORING 00:09:06 PAUSED Resume

admin Evaluator

Description:

Allows administrators to adjust detection sensitivity (such as traffic rate limits and port scan thresholds) and enter API keys for external threat lookups.

System Flow (Final Flow)



Date Submitted:	Sept 1, 2026	
Teacher's Feedback	 _____ _____	